Experiment No. 04

Aim: Installation of Hadoop on windows with cloudera.

Bloom's Taxonomy Level: L3 – Apply

Justification: Students execute Hadoop commands to verify the installation, list HDFS directories, create directories, and perform basic file operations.

Objectives

1. To install and configure Hadoop using Cloudera QuickStart VM on Windows.
2. To verify Hadoop installation using basic commands.
3. To perform basic HDFS directory operations.
4. To create and display a text file using Linux commands.

Theory:

Hadoop is an open-source framework designed to store and process large volumes of data across distributed computing systems. It enables the processing of large datasets by dividing the work among multiple computers. Hadoop consists of important components such as Hadoop Distributed File System (HDFS) for distributed data storage and MapReduce for parallel data processing.

Cloudera QuickStart VM provides a preconfigured environment containing Hadoop and related big data tools. In this experiment, Oracle VirtualBox is used to run the Cloudera QuickStart Virtual Machine on a Windows operating system. After starting the virtual machine, Hadoop commands are executed through the Linux terminal to verify the Hadoop installation, view HDFS directories, create directories, and perform basic file operations.

This experiment helps students understand the basic installation setup and working of Hadoop in a virtualized environment and provides practical knowledge of distributed storage and big data processing.

Applications of Cloudera

Cloudera is a data platform that supports the storage, management, and processing of large datasets using big data technologies such as Hadoop.

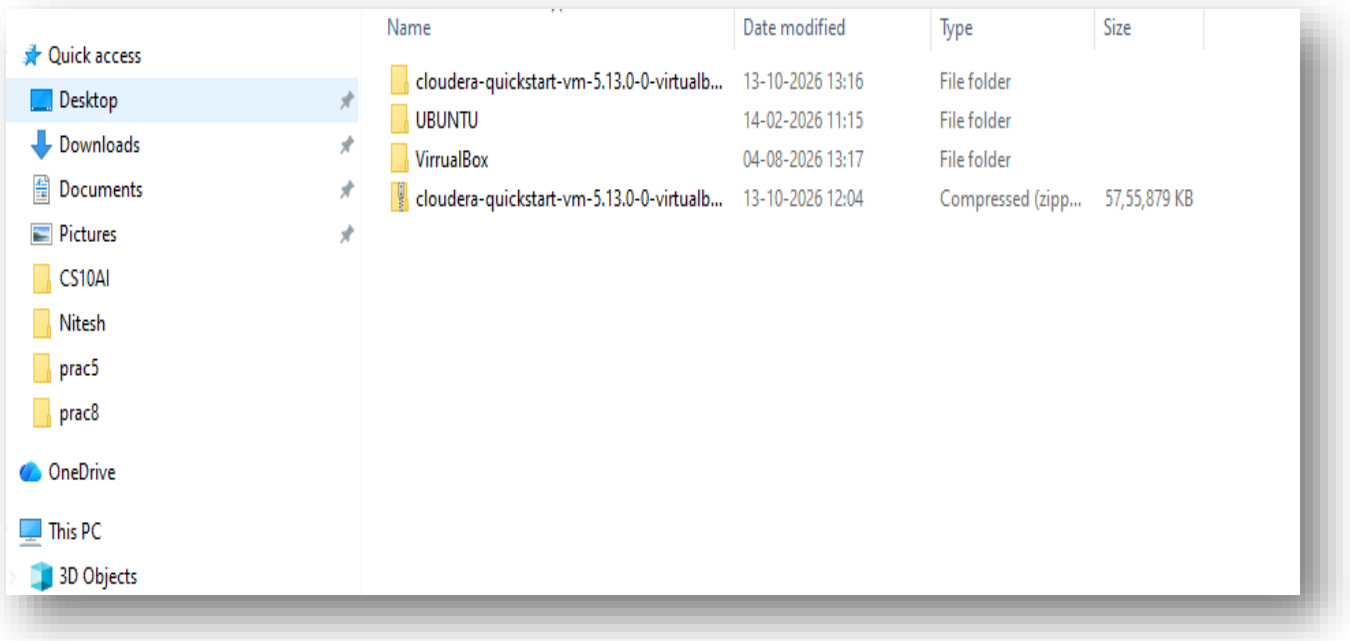
1. Big Data Management: Stores and manages large volumes of data.

2. Data Analytics: Helps analyze large datasets to identify useful patterns and insights.
3. Data Warehousing: Supports the organization and analysis of data for reporting.
4. Machine Learning: Supports data processing workflows used in machine learning.
5. Business Intelligence: Helps organizations generate reports and make data-driven decisions.
6. Log and Event Processing: Analyzes system logs and event data.
7. Data Integration: Helps manage data collected from multiple sources.

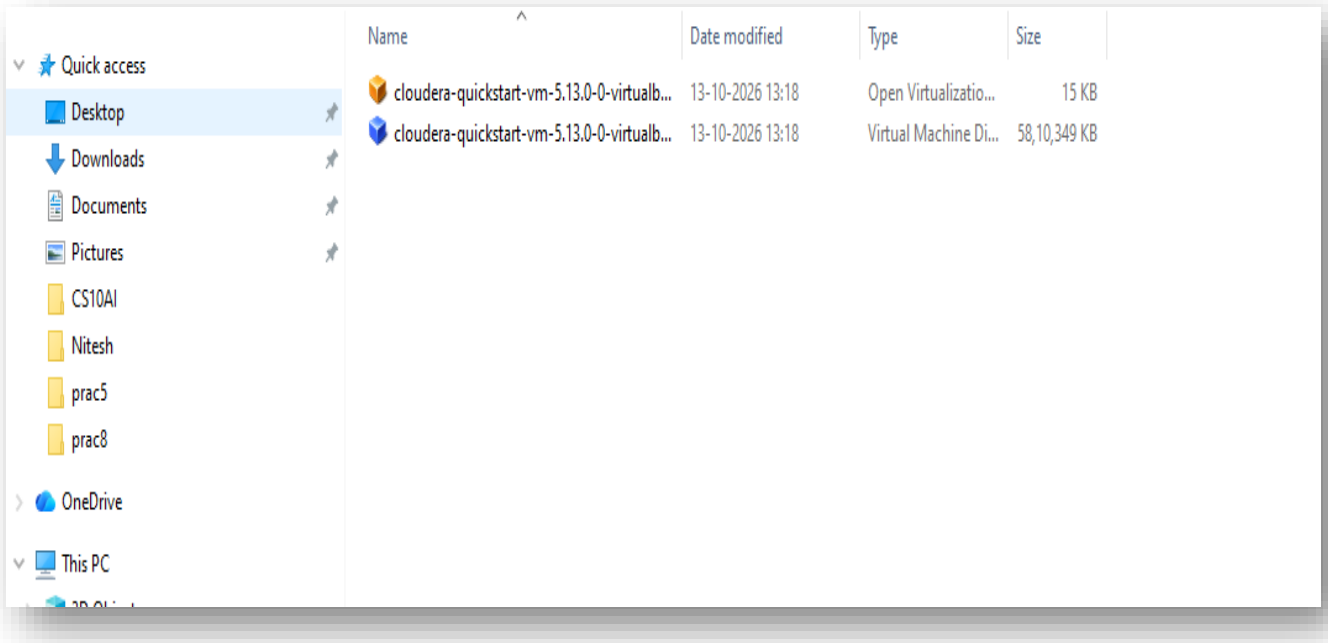
Advantages of Cloudera

1. Integrated Platform: Provides tools for managing and processing big data.
2. Scalability: Supports expansion as data volumes and processing requirements increase.
3. Security: Provides capabilities for data access control and security management.
4. Data Management: Supports data storage, governance, and administration.
5. Analytics Support: Enables data analysis using supported big data tools.
6. Enterprise Support: Offers enterprise-oriented management and support capabilities.

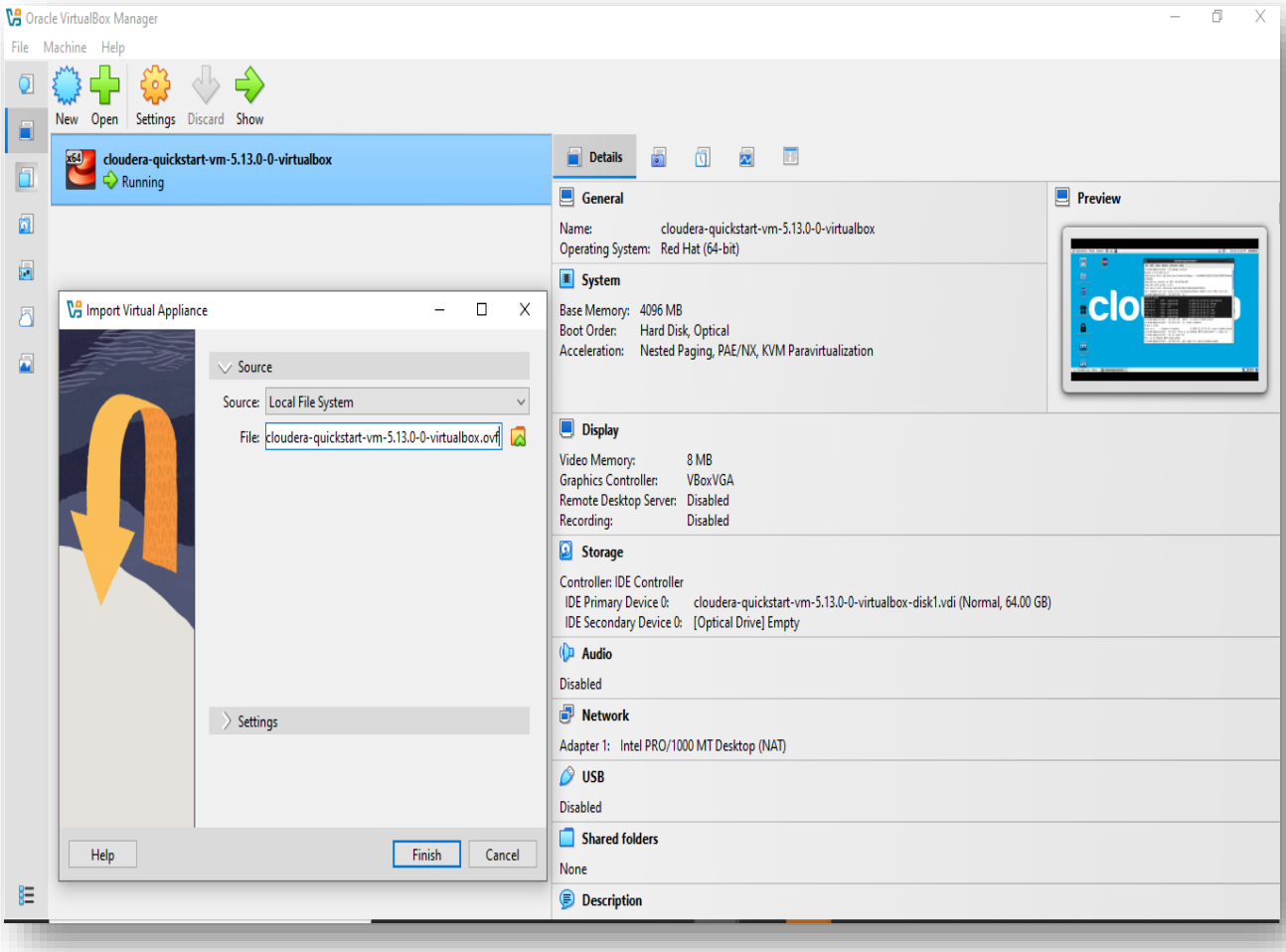
1. Installation and Setup of Cloudera QuickStart VM using Oracle VirtualBox.



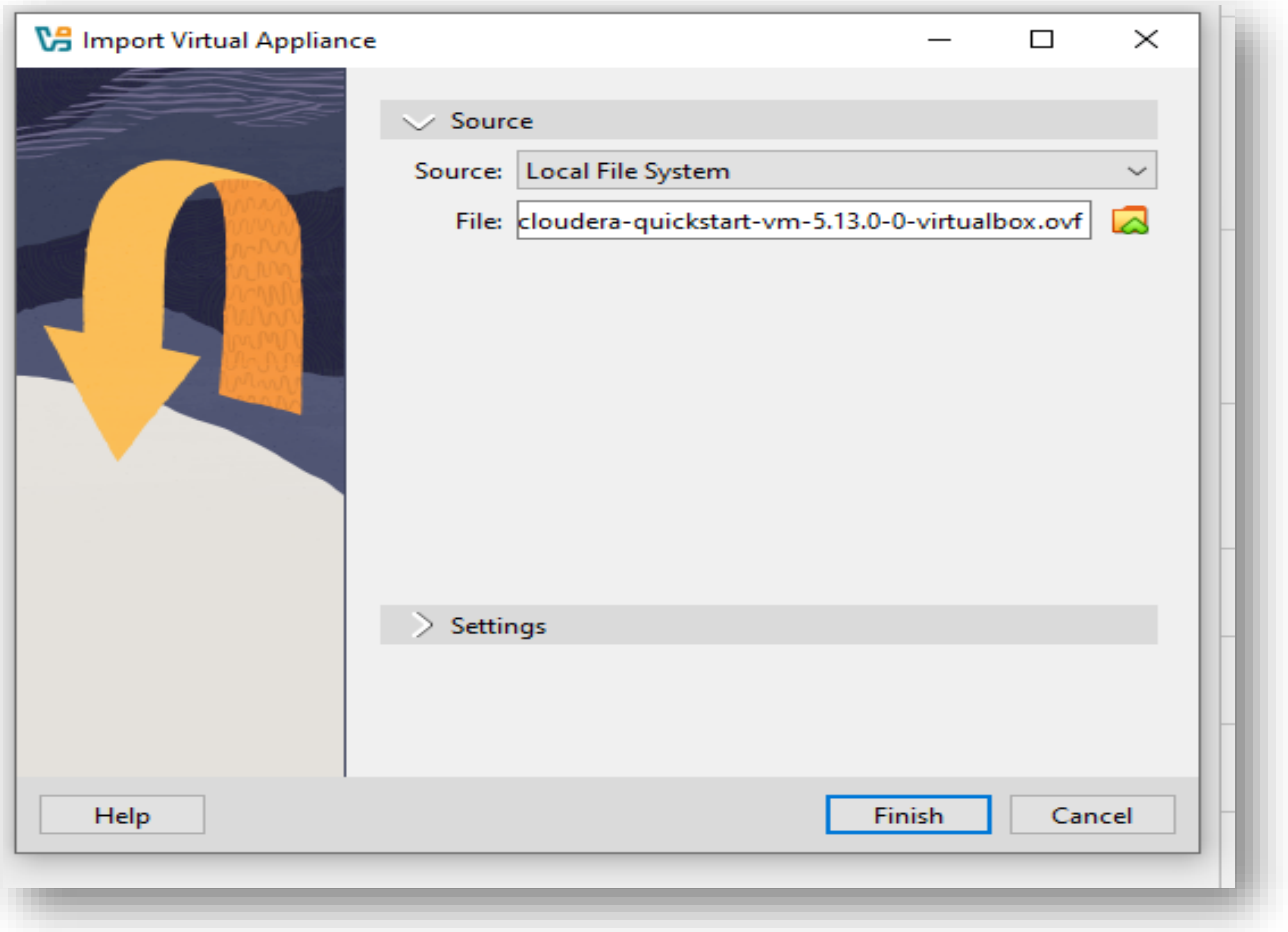
2. Importing Cloudera QuickStart Virtual Appliance into Oracle VirtualBox



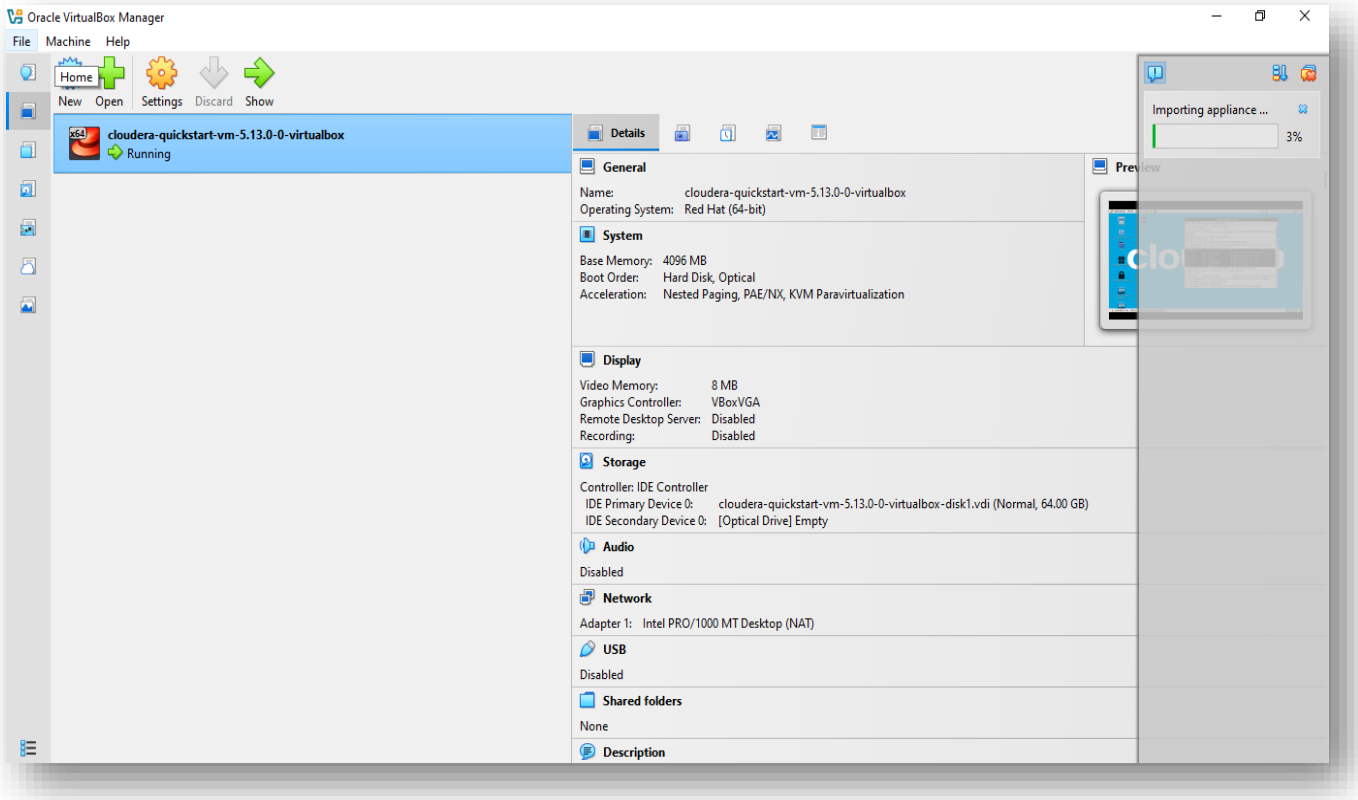
3. Configuring Virtual Machine Settings and Completing the Import Process

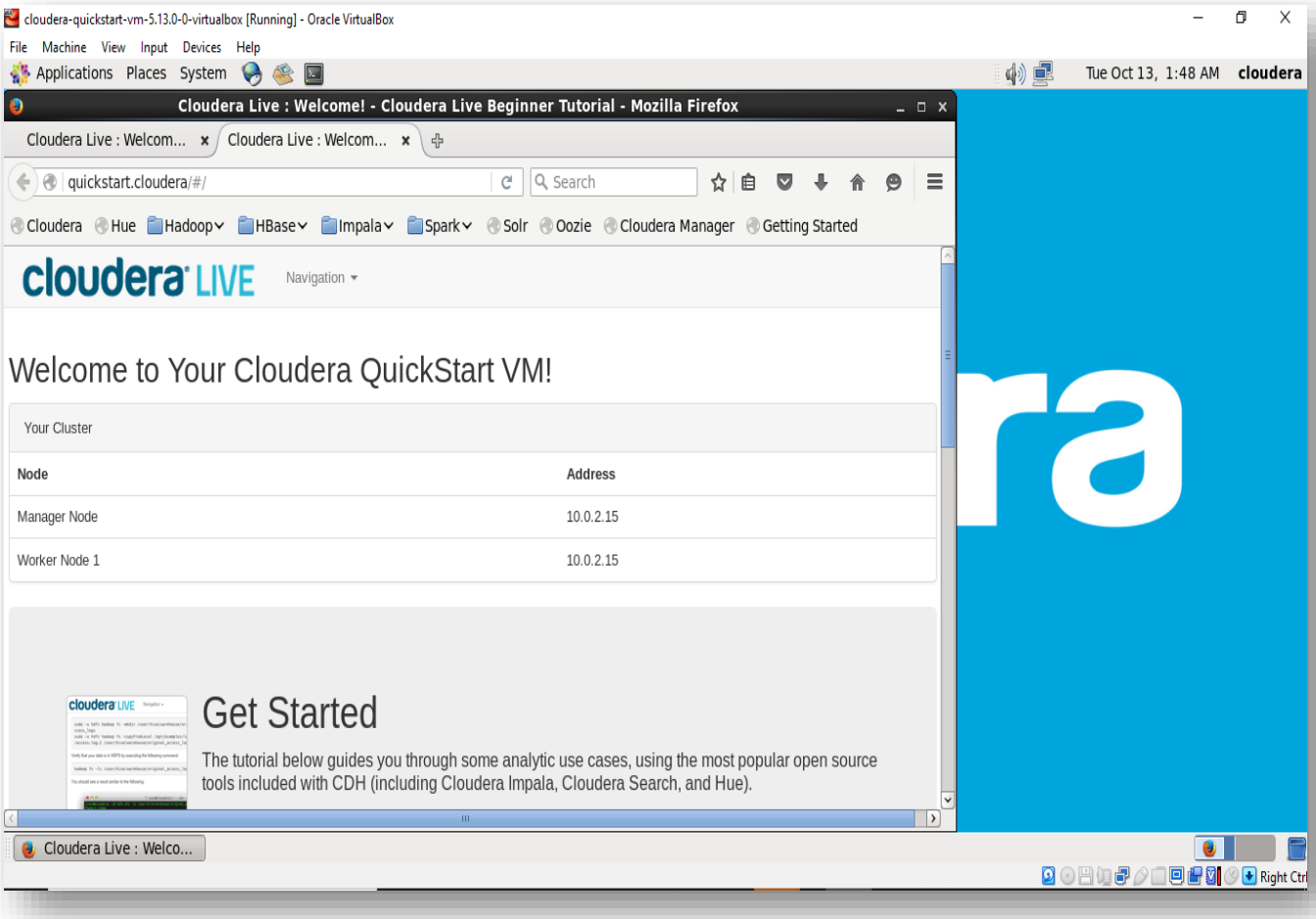


4. Configuring Virtual Machine Settings and Completing the Import Process

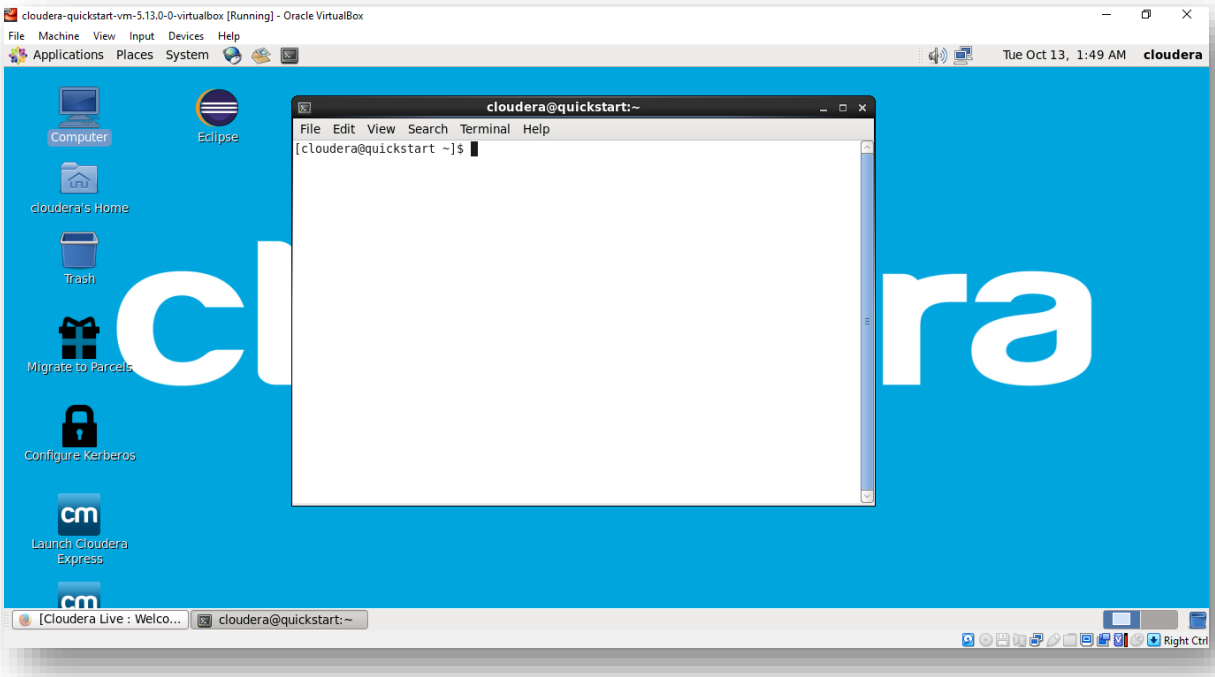


5. Launching the Cloudera Live Environment

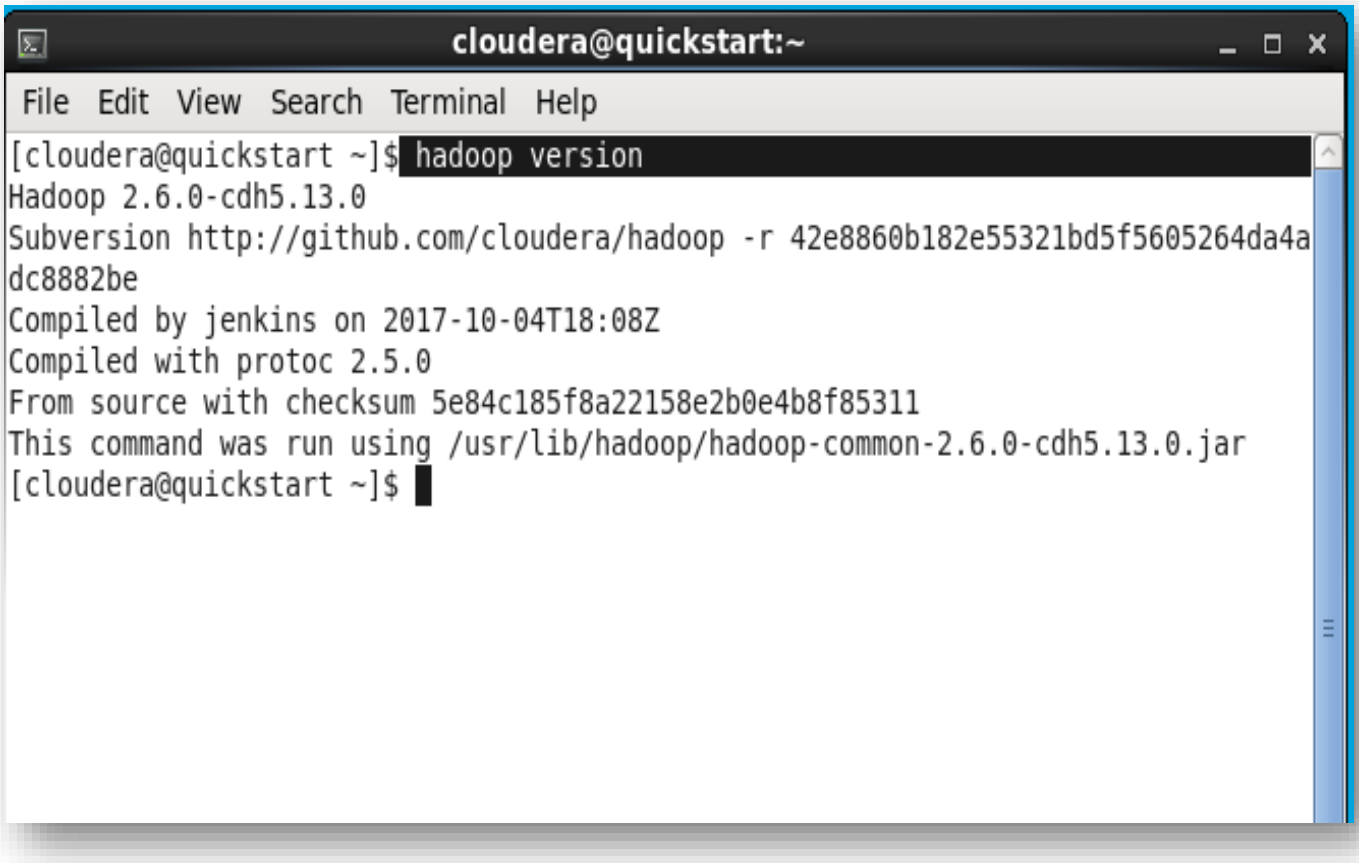




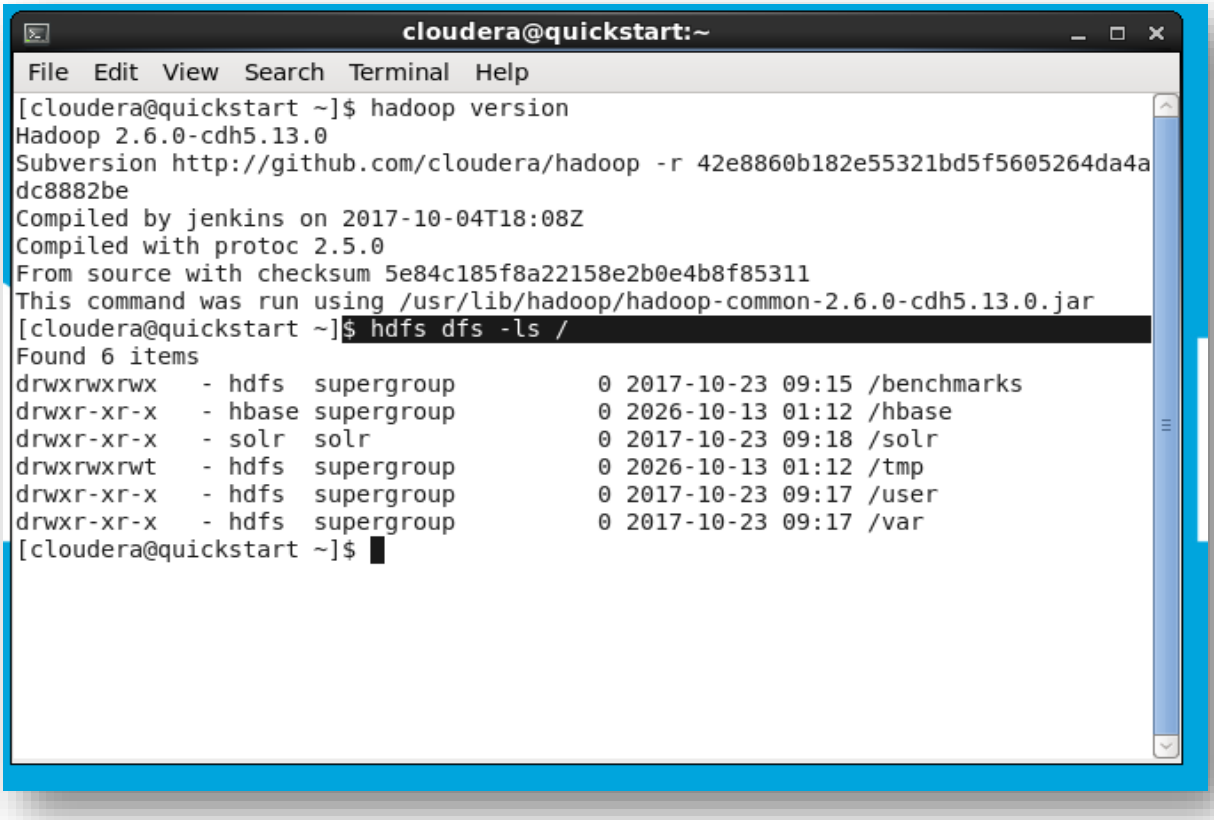
6. Open the Terminal.



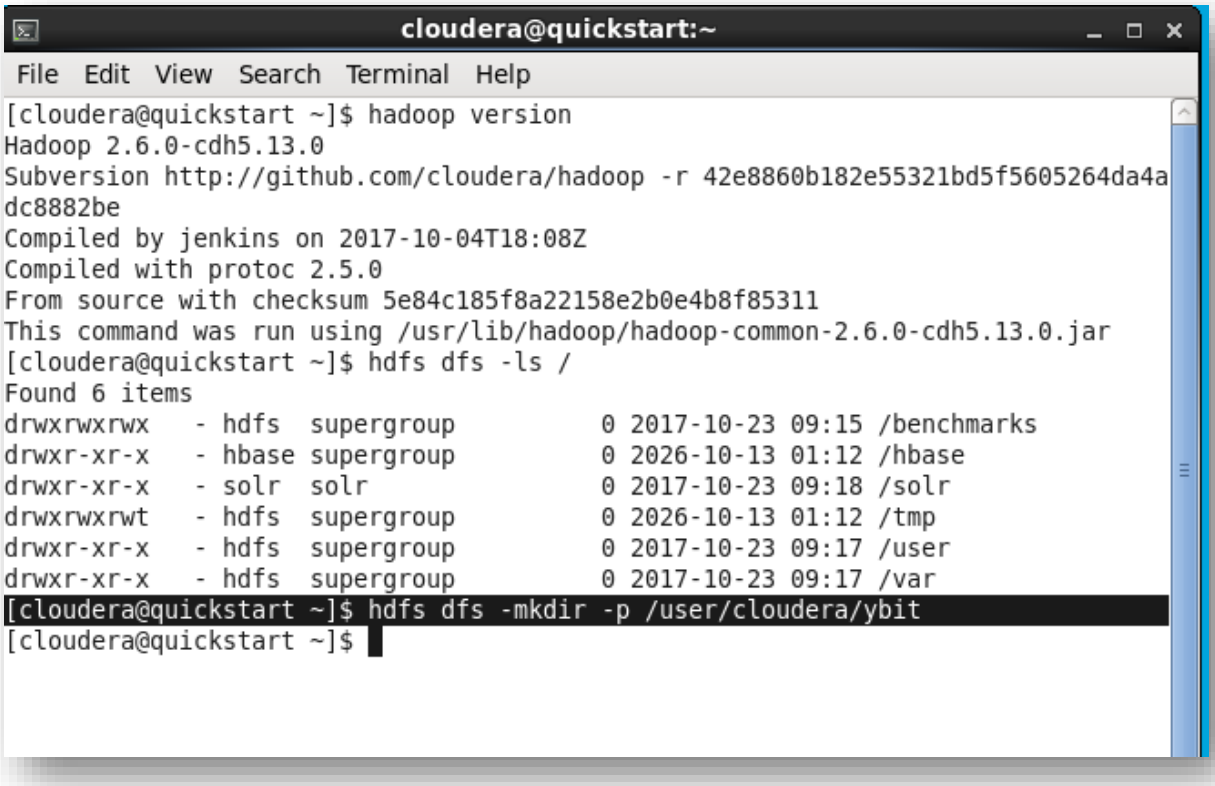
7. Check Hadoop version using Hadoop version.



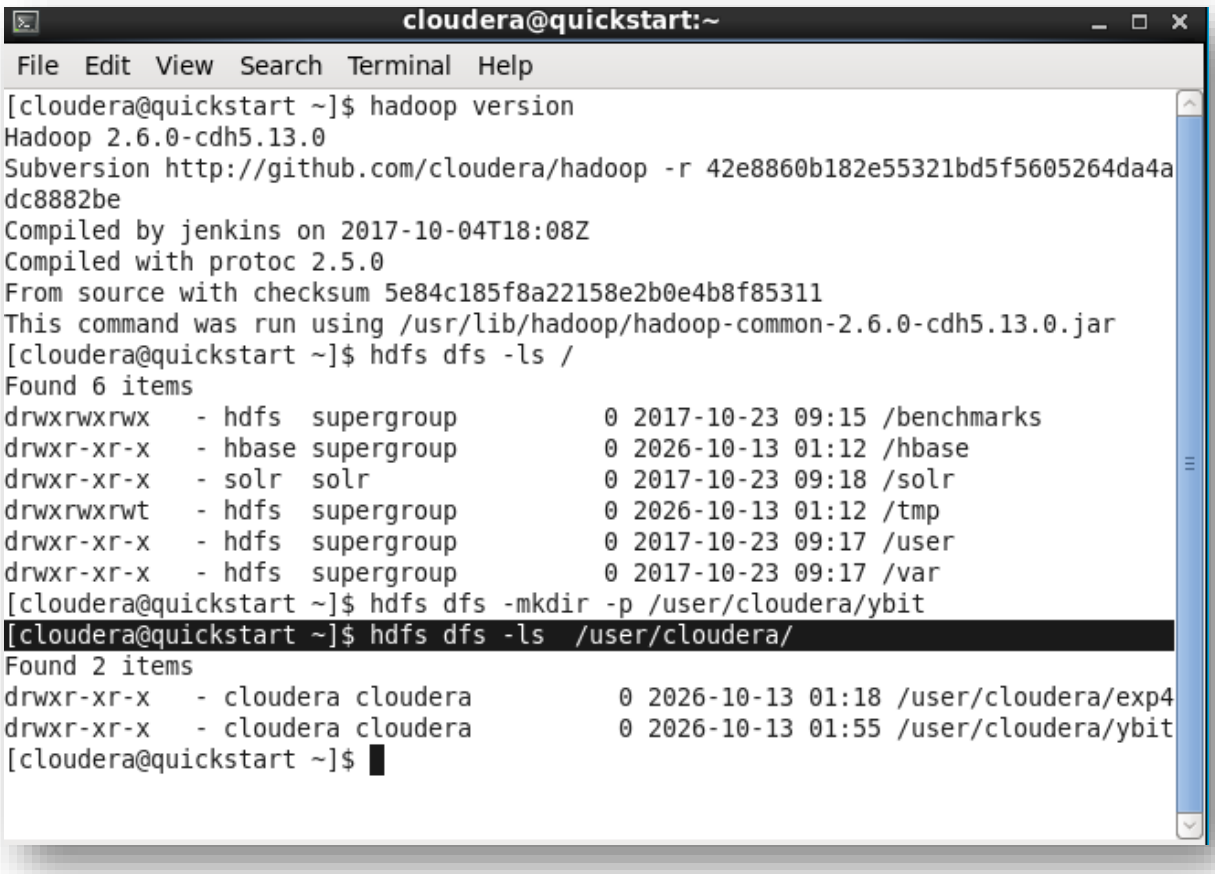
8. Display HDFS directories using hdfs dfs -ls /.



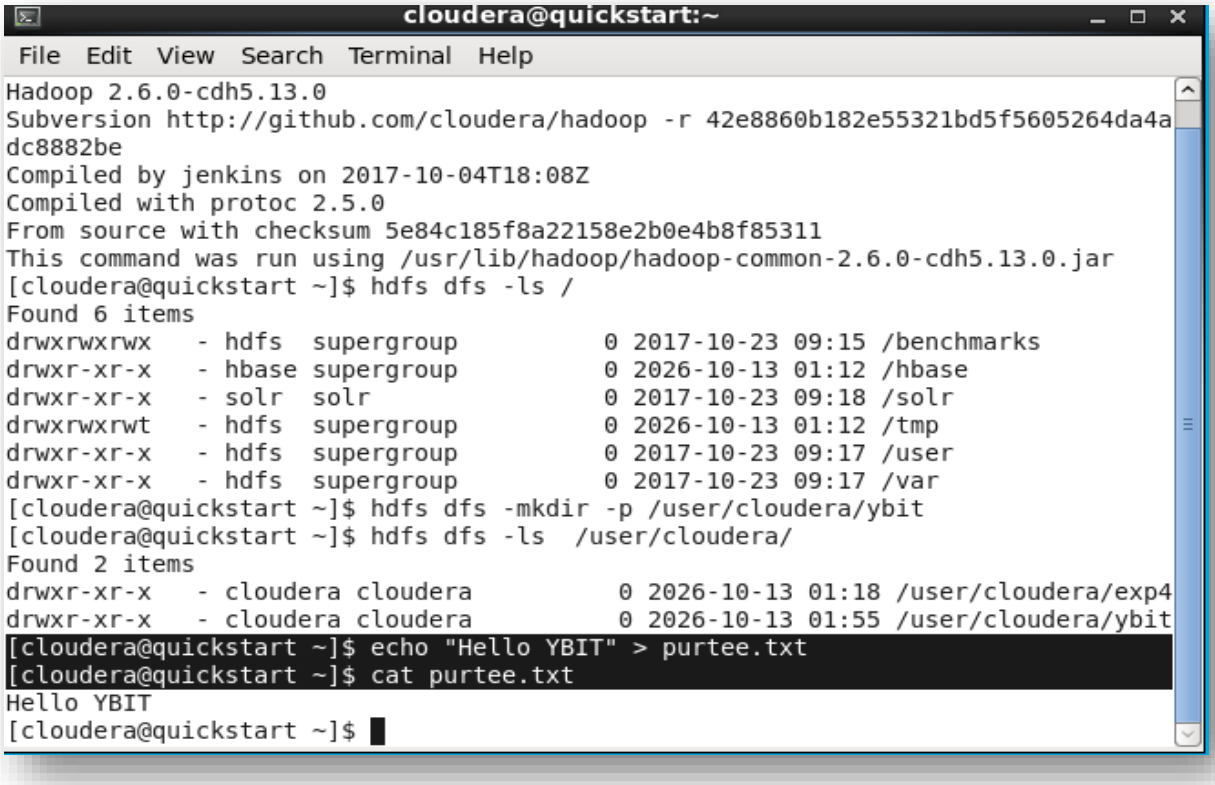
9. Create a directory using hdfs dfs -mkdir -p /user/cloudera/ybit.



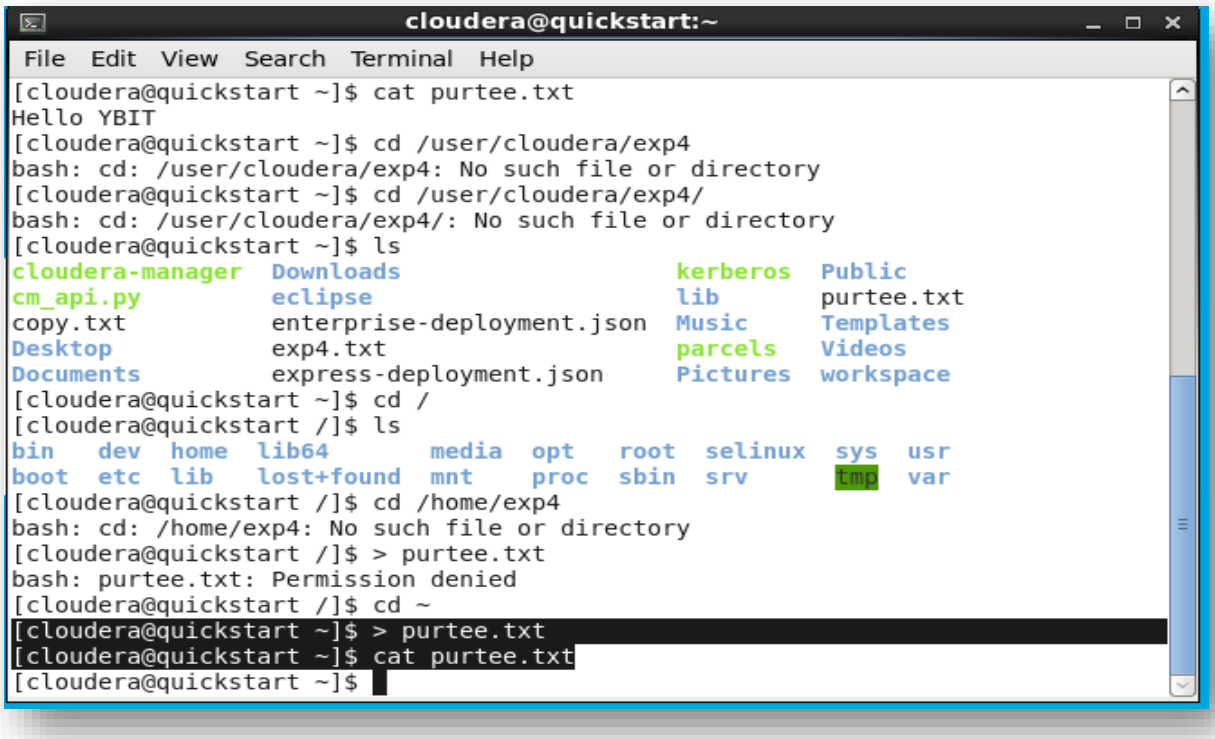
10. Display the created directory using hdfs dfs -ls /user/cloudera/.



11. Create a text file using the echo command.



12. Display file contents using the cat command.



Conclusion: Thus, Hadoop was successfully verified using Cloudera QuickStart VM on Windows with Oracle VirtualBox. Basic Hadoop commands were executed to check the Hadoop version, list HDFS directories, create a directory, and perform basic file operations.